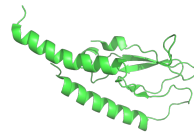




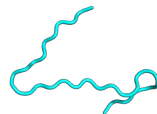
User

Please help me optimize the lead peptide to bind better to the uploaded protein.

Add files



GLP-1 receptor
extracellular domain
PDB id: 4ZGM



Glucagon
peptide

Orchestrate a plan

[ANSWER]

I'll help you optimize the peptide `HSQGTFTSDYSKYLDSSRAQDFVQWLMNT` to improve its binding to the GLP-1 receptor (from the uploaded 4ZGM structure). Let me create a plan and execute it step by step.

Plan:

1. First, I need to understand the current peptide-protein interaction by analyzing the existing complex
2. Use knowledge from literature and structural insights to guide peptide optimization
3. Generate optimized peptide variants based on structural and sequence analysis
4. Evaluate the binding quality of optimized peptides through docking
5. Select the best 2+ peptides that show improved binding over the original

Let me start by getting more information about the protein target and existing peptide knowledge.



Retrieve Agent

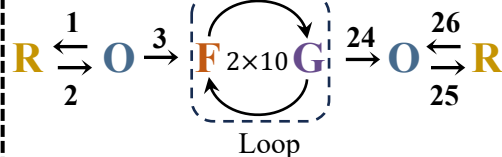


Forge Agent



Gauge Agent

Awaiting execution



a



Orchestrate Agent

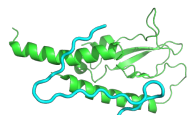
Summary of Content from **R**
Propose peptide design scheme
Propose a work plan to **F** and **G**
Let **R** verify the variants given by **G**

b

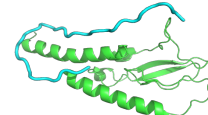


Retrieve Agent

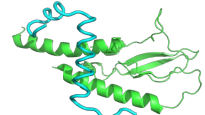
Deep Search **Retrieve and Download**
from literature *from UniProt and E-TSN*
Based on Network Resources
Based on Self-build Database
[TOOL_CALL] TargetDiscovery-get_Standardized_Target_Name
{"target_name": "GLP-1 receptor"}
[TOOL_CALL] Pubmed_Search
{"query": "GLP-1 receptor semaglutide peptide optimization binding affinity"}



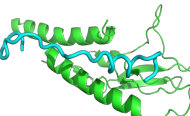
Glucagon Docking
Conformation



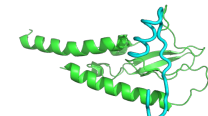
Semaglutide Docking
Conformation



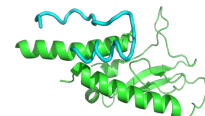
Tirzepatide Docking
Conformation



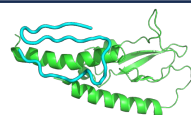
Peptide(a) Docking
Conformation



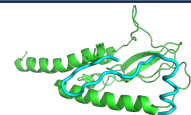
Peptide(b) Docking
Conformation



Peptide(b) Docking
Conformation



Peptide(d) Docking
Conformation



Peptide(e) Docking
Conformation

Requirements
and Plans



Optimized
Peptide
Variants
and
Rankings



Reference

Secondary
Candidate

Primary
Candidate

c



Forge Agent

N epoch



Gauge Agent

Lead Optimization from **O** and **R**

Based on Protein Structure and Sequences
Based on the Literature Knowledge
Based on Reference Ligand Sequences
Based on Deep Learning Models

Optimized Peptide Sequence
Reference Peptide Sequence
Protein Structure Or box(opt.)

Protein-Peptide Docking from **F**

Based on Computer Software
Based on Neural Network Models
Based on Large Language Models
Based on Machine Learning Models
Based on Opensource Webs

De duplication;
clustering and filtering;
Ranking base on Score.
Virtual Screened
Docking
Conformation

Name	Sequences	Score
Glucagon	HSQGTFTSDYSKYLDSSRAQDFVQWLMNT	-2.61
Semaglutide	HAEGTFTSDVSSYLEGAAKEFIAWLVRGRG	-2.60
Tirzepatide	YAEGTFTSDYSIALDKIAQKAFVQWLIAGGPSSGAPPPS	-2.41
Peptide(a)	HSQGTFTSDYVKYLDSSRAQDLVQWLMNT	-2.42
Peptide(b)	HLQGTFTSDYSKYLDSSRAQDFVQWLHNT	-2.46
Peptide(c)	HSQGTGTSFYSKYLDSSRAQDFVQWLMNT	-2.60
Peptide(d)	HSQGTFTSDYSVYLDSSRRFQDFVQWLMNT	-2.66
Peptide(e)	YDQGTFTSDYSKYLDSSRAQDFVQWLMNT	-2.73